MINI PROJECT REPORT

GENERAL SIGNAL GENERATOR

MATLAB GUI Application for Piecewise Signal Creation

SUPPORTED OPERATIONS

Amplitude Scale | Time Reversal | Time Shift | Expand | Compress

PROJECT REPORT

- Interactive Region-by-Region Signal Definition
- DC • Ramp • Polynomial • Exponential • Sinusoidal
- Time Domain Transformations & Visualization

Developed in MATLAB (GUIDE)

Signals & Systems Mini Project

PREPARED BY

Name	ID Number
Amr Mahmoud Moussa	23010637
Yasen Mohamed Ahmed Salhin	23010962

1. Introduction

This project aims to design and implement a **general signal generator** using MATLAB with a graphical user interface (GUI).

The system allows the user to:

- Define a signal over a time interval
- Divide the signal into multiple regions using breakpoints
- Assign different signal types to each region
- Apply operations such as scaling and time transformations
- Visualize the final signal

This project demonstrates key concepts in:

- Signal representation
- Time-domain transformations
- MATLAB GUI development (GUIDE)

2. Objectives

The main objectives of this project are:

- Generate signals with multiple regions
- Support different signal types:
    - DC
    - Ramp
    - Polynomial
    - Exponential
    - Sinusoidal
- Allow user-defined parameters
- Apply signal operations:
    - Amplitude scaling
    - Time reversal
    - Time shifting
    - Expansion
    - Compression
- Display signals graphically

3. System Design

3.1 GUI Layout

The interface is divided into three main sections:

Input Section

- Sampling frequency
- Time range (start & end)
- Breakpoints

Plot Area

- Displays generated and modified signals

Operations Section

- Amplitude scaling
- Time reversal
- Time shifting
- Expansion
- Compression

3.2 Signal Generation Logic

The signal is constructed as follows:

1. The time vector is generated:

$$t = t_{start} : 1/f_s : t_{end}$$

2. The time is divided into regions using breakpoints:

$$regions = [t_{start},\ bp_1,\ bp_2,\ \dots,\ t_{end}]$$

3. For each region:

- User selects signal type via popup
- User enters parameters via dialog
- Signal is computed only for that interval

3.3 Supported Signal Types

1. DC Signal

$$y(t) = A$$

2. Ramp Signal

$$y(t) = mt + c$$

3. Polynomial Signal

$$y(t) = At^p + c$$

4. Exponential Signal

$$\boldsymbol{y(t) = Ae^{at}}$$

5. Sinusoidal Signal

$$\boldsymbol{y(t) = A\sin(2\pi ft + \phi)}$$

---

4. Signal Operations

After generating the signal, the user can apply:

4.1 Amplitude Scaling

$$\boldsymbol{y(t) \to k \cdot y(t)}$$

4.2 Time Reversal

$$\boldsymbol{t \to -t}$$

4.3 Time Shift

$$\boldsymbol{t \to t - t_0}$$

4.4 Expansion

$$\boldsymbol{t \to \frac{t}{a}}$$

4.5 Compression

$$\boldsymbol{t \to a\,t}$$

5. Implementation Details

- **Language:** MATLAB
- **GUI Tool:** GUIDE
- **Key Functions Used:**
  - **inputdlg()** → for parameter input
  - **listdlg()** → for selecting signal type
  - **plot()** → for visualization
  - **guidata()** → for storing data

Important Design Decision

Instead of using fixed GUI inputs for each region, the system uses:

- Interactive dialogs
- This makes the system more flexible and scalable

---

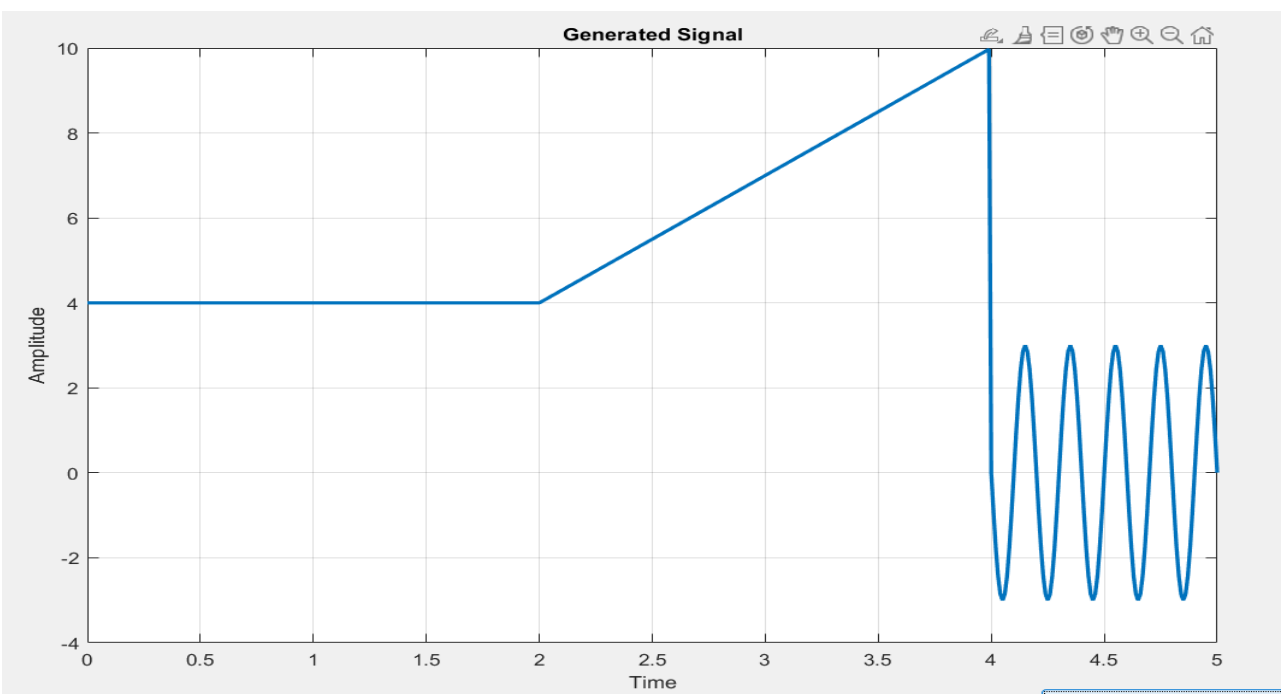
6. Example Run

Inputs:

- Sampling frequency = 100 Hz
- Time range = 0 to 5
- Breakpoints = [2, 4]

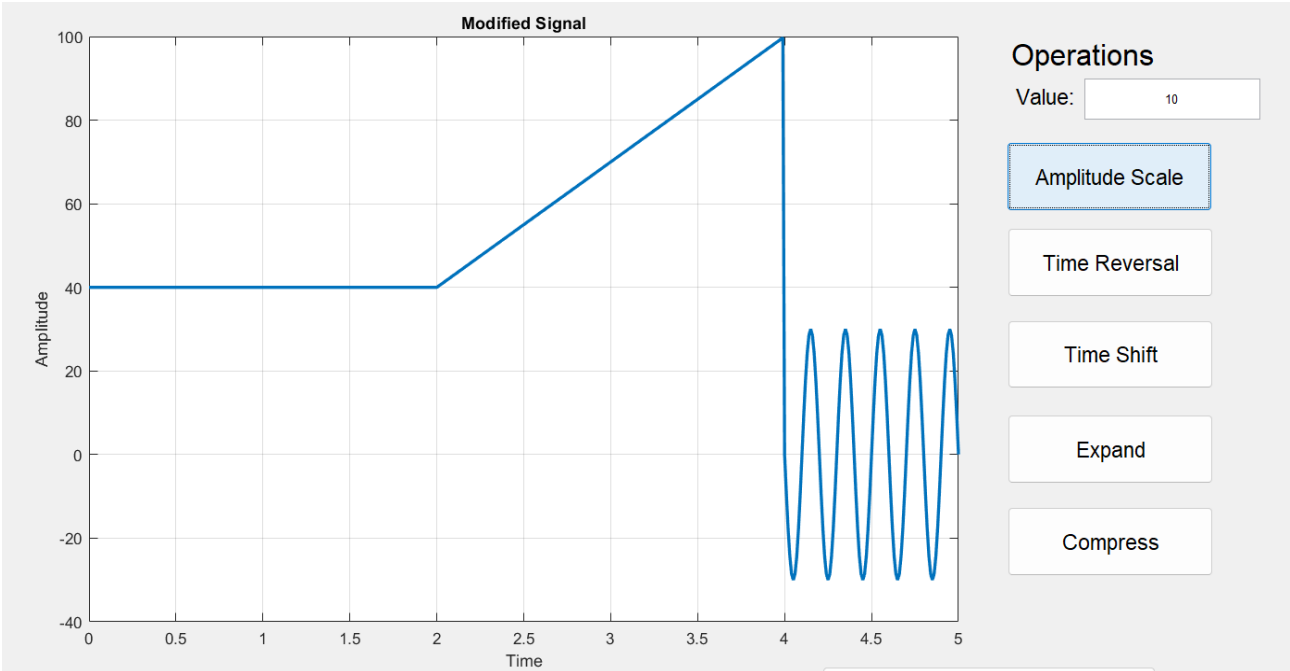
Regions:

1. Region 1 (0 → 2): DC signal
2. Region 2 (2 → 4): Ramp signal
3. Region 3 (4 → 5): Sinusoidal signal

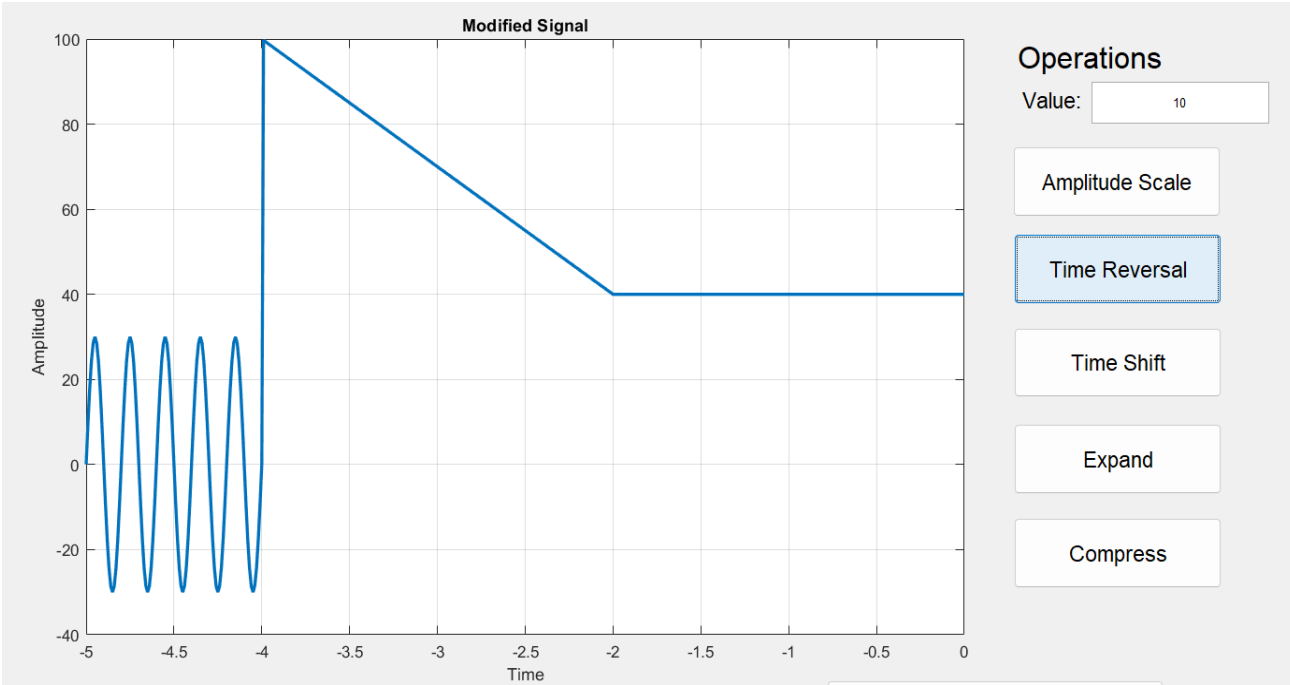


Operations:

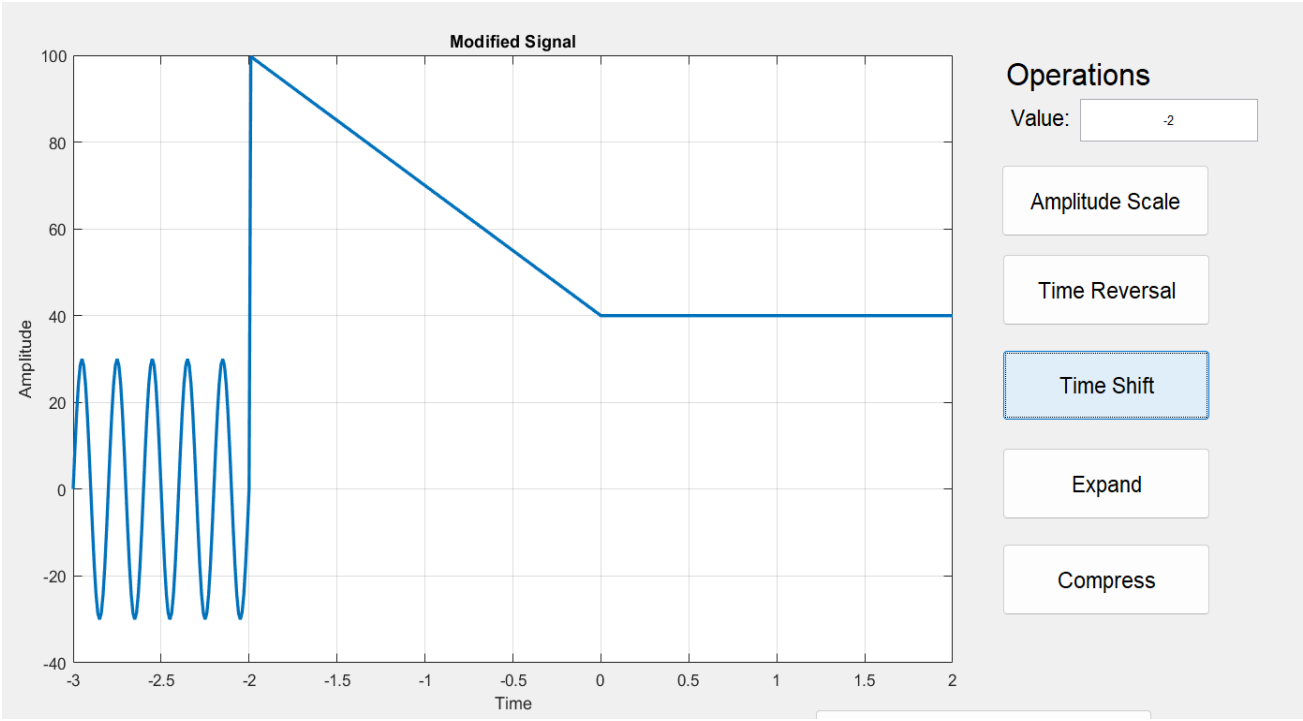
Amplitude Scaling: multiplies the signal by a constant value



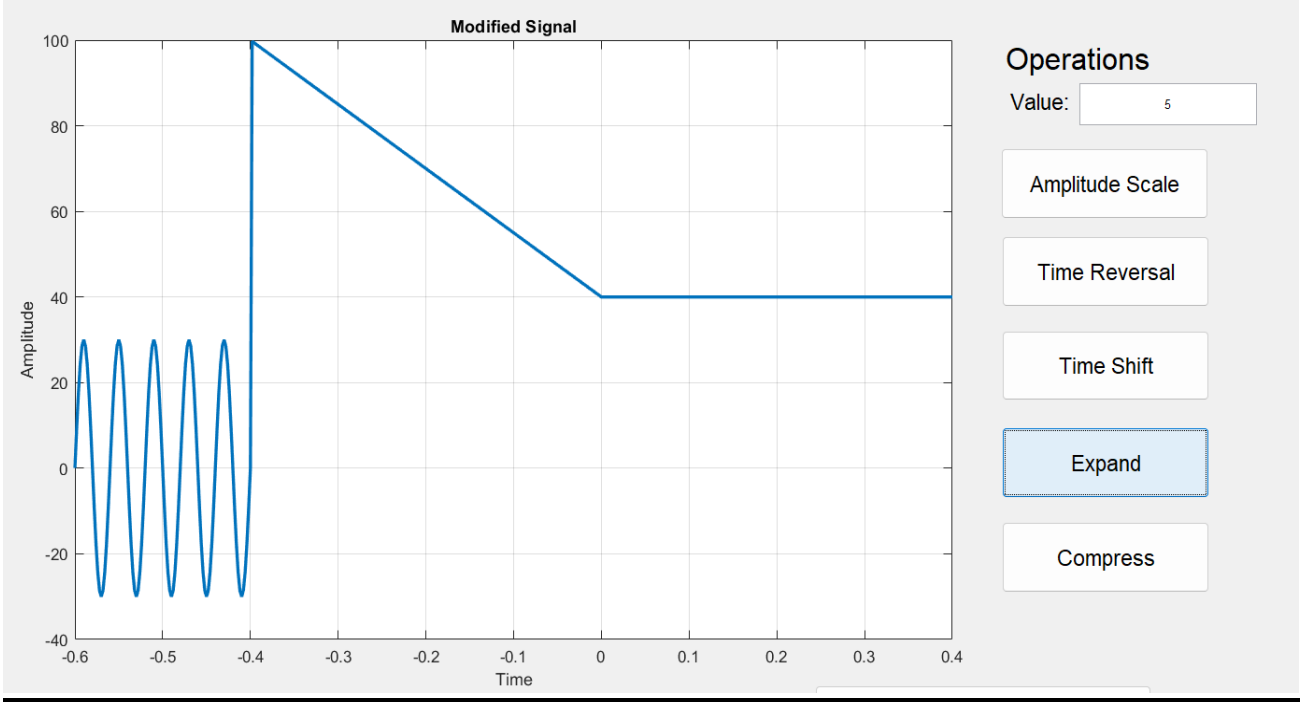
Time Reversal: flips the signal around the vertical axis



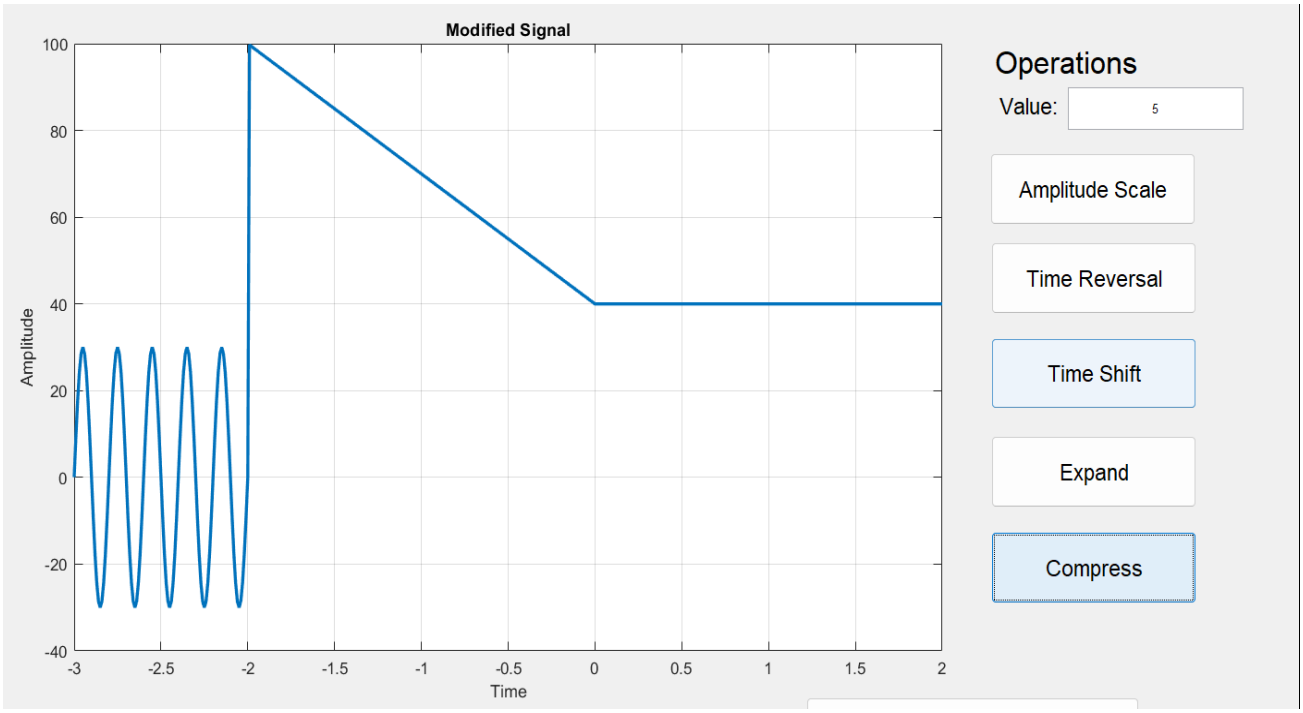
Time Shift: moves the signal left or right



Expansion: stretches the signal in time



Compression: compresses the signal in time



---

7. Conclusion

The project successfully implemented a **general-purpose signal generator** using MATLAB GUI.

It demonstrates:

- Signal modeling
- Time-domain transformations
- Interactive GUI programming

The system is flexible and can be extended to support more signal types and operations.